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Foundational Focus: Basics of Ethernet

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Introduction

A local area network (LAN) provides a path of communication, allowing the delivery of packets of data, voice, or video originating from the sender (logical source address) to the receiver (logical destination address). The most common LAN used is Ethernet.

Ethernet Standards

The Ethernet standard is IEEE 802.3. The IEEE standards are updated periodically to improve the operation and speed of the Ethernet. Each update adds a project amendment suffix to the end of the specified Ethernet standard IEEE 802.3. The suffix is identified by alpha characters, noting a change to the original standard.

Each project amendment has a name identifying the supported speed, the signal type, and cable specifications. The earliest standards called for the use of coaxial cables and were known as 10BASE-5 and 10BASE-2. Today's networks use either unshielded twisted pair (UTP) or fiber cable. The use of UTP cable was first introduced in the 1980s with project amendment IEEE 802.3i.

10BASE-T: (IEEE802.3i) and 10BASE-F

Figure 1: Translating the Ethernet Standard

- **10** identifies speed at 10 Mbps; definition 10 Megabits per second (megabit is 1 million bits).
- **BASE** identifies the signal as baseband, allowing a single signal on a pair at a time.
- **T** identifies the type of cable or length of cable supported for this standard. It also identifies the type of cable for the network as twisted pair. This allows support for unshielded and shielded twisted pair cable at 100 meters maximum length. Although both varieties of twisted pair cable are supported, UTP (unshielded twisted pair) cable became more popular. UTP specification was Category (Cat) 3, which was in the structured wiring of buildings at the time.
- **F** identifies in 10BASE-F, the support for multi-mode fiber cabling. The maximum length is 2000 meters.

Use any of the varieties of Ethernet standard specifications and apply the previous information to learn the speed of the network, the signal type (Ethernet is always baseband for LANs), and supported cable information, such as the type of cable or the length of cable.

Each standard has a maximum cable length based on attenuation. Attenuation is the strength of the signal lost while travelling a distance on the cable. Fiber has much less attenuation loss; therefore, it supports longer cable distances than UTP.

Current Ethernet standards identify multiple categories of UTP cable (Cat. 3–7a); however, all standards allow a maximum length of 100 meters through 1000BASE-T. New standards identifying higher speeds may specify a different maximum length. All of the standards provide for use of fiber cabling; however, there are two basic types of fiber: multi-mode using light emitting diode (LED) to create the signal, and single-mode fiber using laser light to create a signal. The supported lengths and type of fiber may vary per standard.

Below are examples of Ethernet standards and how to translate the meaning.

10BASE-T (10BASE-Ti), 10BASE-F (IEEE 802.3j)

- **F** identifies multi-mode fiber cable.
- **T** identifies twisted pair, Cat. 3.

100BASE-TX, 100BASE-FX (IEEE 802.3u or Fast Ethernet)

- **FX** identifies multi-mode fiber cable.
- **T** identifies twisted pair, Cat. 5.
- **X** identifies full-duplex support in addition to half-duplex.

1000BASE-SX, 1000BASE-LX, 1000BASE-LH (IEEE 802.3z or Gig Ethernet)

- **SX, LX, and LH** support multi-mode and single-mode fiber cable.
- **SX** (Short Wave) supports multi-mode fiber with full-duplex support.
- **LX** (Long Wave) supports multi-mode and single-mode fiber with full-duplex support.
- **LH** (Long Haul) supports single-mode fiber only.

1000BASE-T (IEEE 802.3ab) 1000 Mbps over twisted-pair (1000x1million bits per second)

- **T** identifies twisted pair cable supporting Cat. 5e UTP with a maximum length of 100 meters.
- **STP** (shielded twisted pair) cable is supported with a maximum length of 25 meters.

10GBASE-ER, 10GBASE-SR, 10GBASE-LR, 10GBASE-SW, 10GBASE-LW, 10GBASE-EW (IEEE 802.3ae)

- **Extended Range (ER), Short Range (SR), Long Range (LR), Short Wave (SW), Long Wave (LW), and Extended Wave (EW)** supports different fiber-cable types and distances. (Not all are listed in this paper.)
- **10GBASE-T (IEEE 802.3an)** supports unshielded twisted pair cabling, Cat. 6a or Cat.7; gigabit is 1 billion bits per second; 10G is 10x1 billion bits per second or 10 billion bits per second. 10GigE only supports full-duplex, point-to-point connections between network switches or routers. Hubs and half-duplex are not supported.
- **40G and 100GBASE-(IEEE 802.3ba)** supports multi-mode fiber and single-mode fiber.

Ethernet Topology

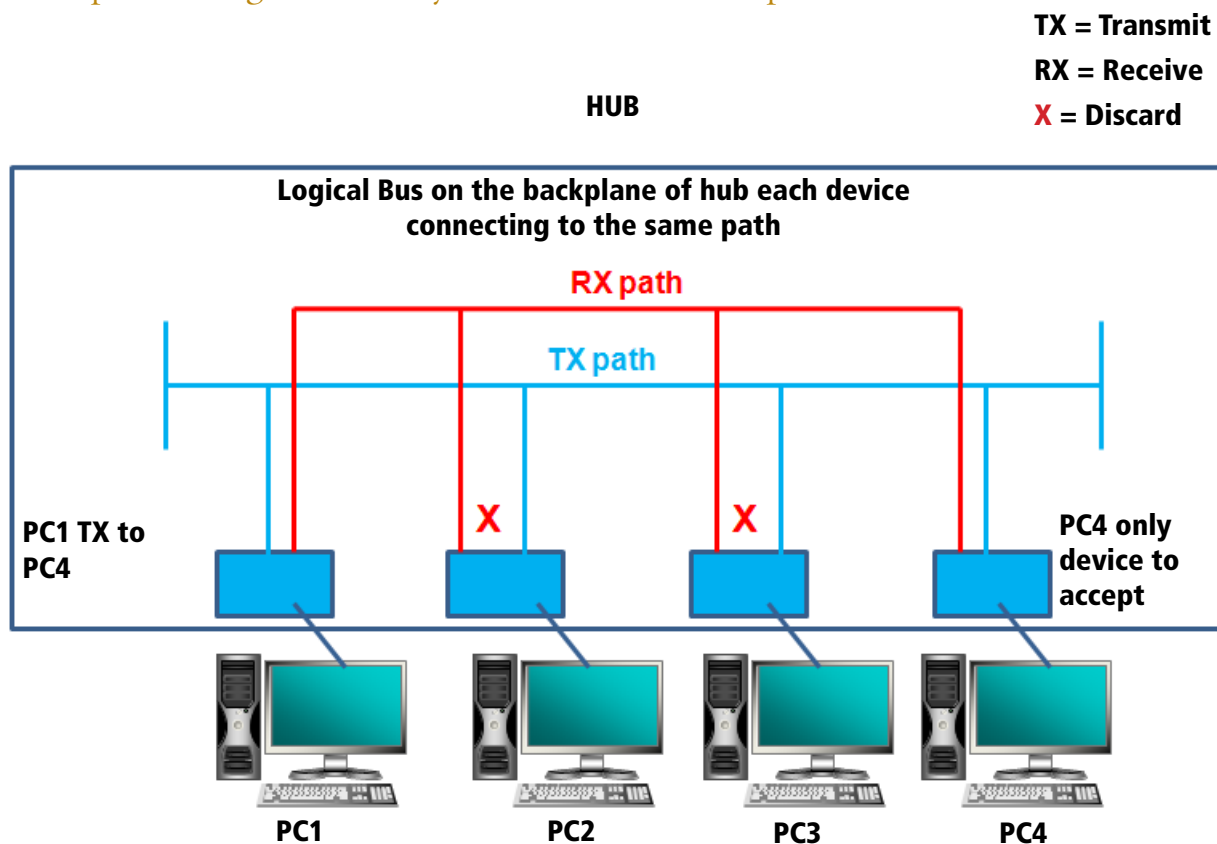
Ethernet IEEE 802.3 is identified as 10Mbps, using a bus topology with CSMA/CD over a shared network path. Half-duplex communication is necessary when all devices are connected to a single common wire, creating a single collision domain. A single collision domain is defined as multiple devices sharing the same network path, and only one is allowed to transmit at a time. If two devices transmit at the same time, a collision of signals occurs, causing both stations to retransmit. The topology was updated in the mid-1980s to a logical bus, physical star with the introduction of hubs and 10BASE-T. The speed or the duplex did not change, remaining half-duplex (see Figure 2).

Logical Bus/Physical Star with CSMA/CD

IEEE 802.3 with CSMA/CD is a LAN access method that allows only a single network device to send data (transmit) at a time; this is known as half-duplex. To transmit, each device must follow a set of rules called CSMA/CD, defined as carrier sense (CS), multiple access (MA) with collision detection (CD).

- **Carrier sense:** Is there a signal on the wire? No, I can transmit after hearing quiet for a pre-determined time, based on the speed of the network. The quiet time required on a 10Mbps half-duplex network is .9 micro seconds. (Note: the faster the network speed, the less quiet time is required.)
- **Multiple access:** When one device transmits, all devices on the same collision domain hear the signal, but only the device that identifies the unique physical address of its network interface card will accept and process the frame. All other devices will identify it as not addressed to them and discard the frame.
- **Collision detection:** The issue with half-duplex communication is that sometimes two devices do hear quiet at the same time and both transmit, causing a collision (of signals). As the two signals collide on the half-duplex (single) path, each device must send a 4-byte "jam signal," making sure all other devices on the shared network receive the signal so they do not transmit. Each then uses the defined back-off algorithm to wait a period of time, based on a random timer, before trying to transmit again. The networks that are required to follow CSMA/CD are 10BASE-5, 10BASE-2, 10BASE-T, and 10BASE-F.

Example of a Logical Bus/Physical Star and Half-Duplex



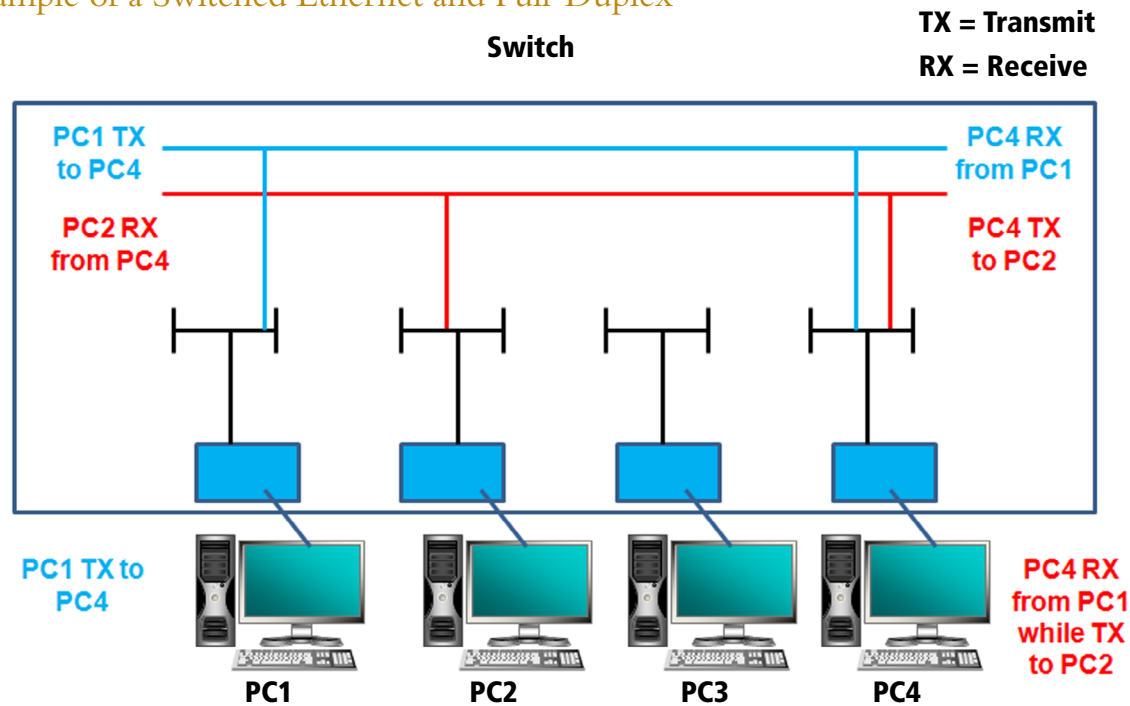
PC1 transmits (TX) to PC4. All devices, PC2, PC3, and PC4, hear the signal. Each compares the physical address of their NIC (network interface card) to that of the message. If it is a match, the device will accept the message and process it. If it is not a match, the device will discard the message. Only PC4 will accept the message because it is a match; all others will be discarded.

Figure 2: Logical Bus/Physical Star and Half-Duplex

Switched Ethernet and Full-Duplex

The 100BASE-TX and 100BASE-FX project amendments, identified full-duplex. Full-duplex becomes usable when the network interface card (NIC) of a device supports full-duplex, and the device is connected to a switch port. A switch port with full-duplex provides transmit and receive pairs available simultaneously to the device, no longer requiring the CSMA/CD functions for sharing a path. Each device has a connection directly to the switch, and each port is equal to a single collision domain. With only one device on a single collision domain, there is no need to listen for quiet, share a path with others, or have collisions. The single device now has a path to itself into the network; there are no other devices to contend or share with. This improves performance and lessens errors.

Example of a Switched Ethernet and Full-Duplex



PC1 transmits (TX) to PC4, but because it is full-duplex and each device has a connection to a single collision domain separate from each other, the switch can identify the destination physical address and forward to only that device; others are unaffected.

Figure 3: Switched Ethernet and Full-Duplex

Ethernet Addresses: A Point-to-Point Address

The Ethernet address is the physical media access control (MAC) address found hard-coded in the NIC by the manufacturer. The address is a 48-bit, 6-byte address used to connect the device to the network. The MAC address is shown as the "physical address" when displayed in the configuration of a device. The address is specified using hexadecimal (base 16) numbering system of 0 through 9 and A through F. Each character is 4 bits; therefore, two characters are 8 bits or 1 byte.

Ethernet addresses complete local communications with other devices within the same logical area of the LAN. In an IP LAN, this logical area is called a subnet. This is a point-to-point address and functions at the OSI Model Data Link layer 2 (L2). Each Ethernet address (MAC) is unique, based on a combination of vendor code (first 3-bytes of the address) that is unique to the manufacturer of the NIC, and a serial number (last 3 bytes of the address) that is unique to that specific vendor code. The vendor code is registered and assigned by the IEEE. If the NIC is replaced by another NIC, the new card will have a different Ethernet address.

The Ethernet MAC address identifies the next device the packet must pass through on the way to the final destination. The switch will read the destination MAC address in the frame and pass the frame to the port identified

in the switch MAC address table. The MAC address table is a list of physical MAC addresses listing the switch port at which the address will be found. The current destination Ethernet MAC address is not changed, it is just passed on to the identified switch port.

Routers forward the Ethernet frame if the destination device is not in the same logical area of the network. Routers may change the destination and source Ethernet MAC addresses as required to pass it forward to the next router or final device. Think of the Ethernet address, in a routing environment, as a relay address, changing at each router on the journey to the final destination device. If the destination device is within the same logical area as the source, the destination Ethernet physical address will not be replaced, as it will find the destination without the help of a router.

00-00-00-22-22-22
Vendor code: First 3 bytes shown as 00-00-00
Serial number: Last 3 bytes shown as 22-22-22

Figure 4: Ethernet, MAC or Physical Address

Logical Address: An End-to-End Address

A logical address is used inside the protocol packet to identify the original source and the final destination of the packet or message. A logical address is assigned to the device automatically by a server or statically by a person. In current networks, the IPv4 address is most often used as the logical address.

An IPv4 logical address is a 32-bit, 4-byte, dotted decimal address that defines the LAN network ID, subnet ID, and host address. This address is considered an end-to-end address because, unlike the Ethernet address, it is not changed by any device and, therefore, always identifies the two communicating devices.

192.0.2.3/24

Figure 5: Logical Address: An End-to-End Address

The **/24** is a prefix that identifies how many bits of the address are network ID and subnet ID. The prefix is a simplified way to denote a subnet mask of 255.255.255.0.

Note: There are currently two versions of TCP/IP: version 4 and version 6. Version 4 has depleted all the available addresses, making it necessary to create a new version to improve the IP functionality and have more IP addresses available. IPv6 does not use the same address format as IPv4.

Ethernet Frame Format

The Ethernet frame format includes an Ethernet header and trailer that “frame” the packet (L3–7 payload). The Ethernet header is 14 bytes and includes information in the following order:

- **Destination MAC address:** First 6 bytes of the header
- **Source MAC address:** Second 6 bytes of the header
- **Ethernet type field:** A 2-byte header following the source MAC address. The type field contains a 2-byte hexadecimal value identifying the payload (L3 – L7) carried in the frame. A common example is the identification of IPv4 as the payload type by “0800”. This is one of many type codes.
- **Data packet:** the payload or message part of the frame known as the packet. The packet is the important message that is delivered to the final destination, which is identified by the logical address in the first header inside the data packet. The data packet carries information from L3 through L7 of the routed/message protocol.

The minimum length of an Ethernet frame is 64 bytes, and the maximum length is 1518 bytes. If a frame is too small, it causes a **Runt** error; if it is too big, it causes a **Giant** error. If the length is not valid, the frame is discarded, and an error of either runt or giant is given. The frame must be retransmitted.

The Ethernet Version II frame format is the frame format; however, there are two other frame formats identified by IEEE that may be used with Ethernet. They are IEEE 802.3 with 802.2 logical link control and IEEE 802.3 with subnetwork access protocol (SNAP). The major difference is that the type field becomes a length field, providing the length of the packet (not the frame). Both use additional bits from the data packet field, shortening that field from a maximum of 1500 bytes to a maximum of between 1497 bytes and 1492 bytes. These extra bytes provide the packet type, since the type field is used for length in these frame styles.

- **Frame Check Sequence (FCS).** A 4-byte trailer that uses the cyclic redundancy check (CRC) mathematical algorithm to validate the accuracy of the data as the frame arrives at each device. The accuracy is verified from the destination MAC address to the end of the data packet. If the frame contents are not valid, the frame will be dropped and must be resent. A CRC error is reported.

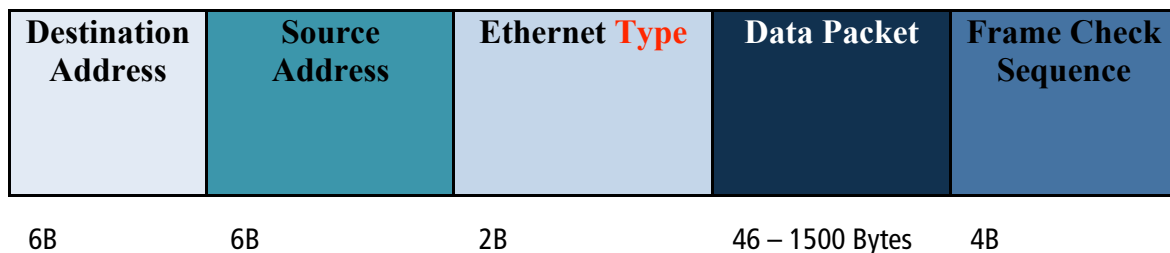


Figure 6: Ethernet Version II Frame Format

Each field length is identified in B = bytes

Understanding the LAN

As you start to learn about networking, remember that communication and the movement of large numbers, whether it is people, cars, mail, or network traffic, have a commonality. Everything you know and use in your daily life can be compared to the way traffic moves.

Understanding the world of networking is easier if you can make an analogy of functions used in daily life. The Post Office is used in this example in comparison to how an Ethernet frame is built, sent, received, and processed.

Email Message to a Co-Worker

Let's go through the steps that occur when you are at work and wish to send a message to your co-worker in the same department.

- You open your email application, enter the email address of your co-worker, and write your message.
- You click **send**, and the message is sent. It is now sitting in your computer waiting for the addressing and frame format.
- When you reach the point of sending the message, your computer matches the email address of your co-worker with their logical address (IP address) by sending a query to the Domain Name Server (DNS), which keeps those records.
- The DNS server responds with the matching IP address, and the computer enters it in the frame as the destination logical address; your logical address is placed as the source logical address.
- Your computer creates the Ethernet frame to surround the Data Packet and guide it from your source computer to the destination computer of your co-worker.
- The **type** field is identified with hexadecimal 0800, identifying IPv4 as the packet type.
- The physical MAC address of your NIC is added to the **source MAC address** field of the Ethernet frame. Your computer checks its own address table of logical and physical pairs of addresses and finds a match for your co-worker. It then adds your co-workers physical MAC address to the **destination MAC address** field of the Ethernet header.
- The frame check sequence CRC algorithm verifies that the bits in the frame arrive to your co-worker correctly. Now your frame is sent to the network switch.
- The switch compares your Ethernet address as the source and verifies it is registered correctly in the switch MAC address table.
- The switch then compares the destination Ethernet MAC address to the switch MAC address table to find a match; a match is found; the table identifies the port connected to your co-worker's computer; and the switch forwards the Ethernet frame to only that port.
- Your co-worker's computer receives the Ethernet frame and takes the following steps:
 - It finds and verifies a match of the destination Ethernet MAC address to the NIC Ethernet address, so this message is for it.
 - The frame length is verified as a legal length.
 - The Frame Check Sequence field value is validated as correct; the Ethernet frame is accepted for processing.

- Your co-worker has received your message. Your co-worker reads the message saying, *"Happy Birthday."* Your message has been received.

A card is sent to a friend

Now let's compare the process of your email message on the network to the way a card is sent to your neighbor using the Post Office. Comparisons to the network are in parenthesis.

1. You both live in the same town and zip code (work in the same department).
2. You write *"Dear Neighbor"* (email address) on the top inside of the card (email/data packet), sign your name and put it in an envelope (Ethernet header/trailer).
3. You add your neighbor's address to the front of the envelope and your address as the return address. (The envelope has two address functions; house number and street are compared to the physical MAC address; and the City, State, and Zip are compared to the logical address.)
4. You deposit the card in the post box on the corner.
5. The mailman (NIC sends the frame to the network) picks up the mail from the box and returns to the Post Office where your card is put with other mail.
6. A mail sorter (the switch) identifies which postman should deliver to the address (destination MAC address) on the card.
7. The card is picked up by the correct postman (sent to the switch port identified in the switch MAC table) and delivered to the address on the envelope (the computer with the matching destination MAC address).
8. Your neighbor identifies that this mail is addressed correctly to him (the NIC verifies MAC address match, legal length validity, and validates FCS; is the message readable).
9. Knowing the mail is for him; he opens the envelope and sees the card is from you saying *"Happy Birthday."* Your card has been received.

Conclusion

A local area network (LAN) provides a path of communication, allowing the delivery of packets of data, voice, or video originating from the sender (logical source address) to the receiver (logical destination address). Ethernet is the most common LAN used. As you start to learn about networking, remember that communication and the movement of large numbers, whether it is people, cars, mail, or network traffic, have a commonality. Everything you know and use in your daily life can be compared to the way traffic moves.

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About the Author

Cindy Kvek was involved in the installation and maintenance of LANs from the early days of networking in Silicon Valley. She is now a consultant and technical instructor specializing in networking with TCP/IP, version 4 and 6. She also enjoys teaching the basic LAN classes, and enjoys sharing her knowledge and experience with students.